

4(a)	<table border="1"> <thead> <tr> <th data-bbox="300 248 456 342">polymer</th><th data-bbox="456 248 954 342">constituent monomer</th><th data-bbox="954 248 1318 342">type of bond between monomers</th></tr> </thead> <tbody> <tr> <td data-bbox="300 342 456 409">amylose</td><td data-bbox="456 342 954 409">glucose</td><td data-bbox="954 342 1318 409" rowspan="2">glycosidic glycosidic ; A glucosidic I details of bond</td></tr> <tr> <td data-bbox="300 409 456 544">cellulose</td><td data-bbox="456 409 954 544">glucose</td></tr> <tr> <td data-bbox="300 544 456 645">collagen</td><td data-bbox="456 544 954 645">amino acid ; R protein (amino acid)</td><td data-bbox="954 544 1318 645">peptide</td></tr> <tr> <td data-bbox="300 645 456 712">DNA</td><td data-bbox="456 645 954 712">DNA nucleotide</td><td data-bbox="954 645 1318 712">phosphodiester ;</td></tr> <tr> <td data-bbox="300 712 456 813">glycogen</td><td data-bbox="456 712 954 813">$(\alpha\text{-})$glucose ; R β-glucose</td><td data-bbox="954 712 1318 813">glycosidic</td></tr> </tbody> </table>	polymer	constituent monomer	type of bond between monomers	amylose	glucose	glycosidic glycosidic ; A glucosidic I details of bond	cellulose	glucose	collagen	amino acid ; R protein (amino acid)	peptide	DNA	DNA nucleotide	phosphodiester ;	glycogen	$(\alpha\text{-})$ glucose ; R β -glucose	glycosidic	4
polymer	constituent monomer	type of bond between monomers																	
amylose	glucose	glycosidic glycosidic ; A glucosidic I details of bond																	
cellulose	glucose																		
collagen	amino acid ; R protein (amino acid)	peptide																	
DNA	DNA nucleotide	phosphodiester ;																	
glycogen	$(\alpha\text{-})$ glucose ; R β -glucose	glycosidic																	
4(b)	<i>two from</i> 1 amylose composed of α-glucose (monomers) <u>and</u> cellulose composed of β-glucose (monomers) ; 2 (amylose) α-1,4 v. (cellulose) β-1,4 (glycosidic) bonds ; R if amylose stated to have α-1,6 bonds I ref. to cellulose also has hydrogen bonds <i>If mp 1 and 2 not stated allow one mark for amylose has α-glycosidic bond and cellulose has β-glycosidic bond</i> 3 (cellulose) adjacent, monomers / glucose(s), rotated through 180° ; AW R ref. to heads / tails 4 amylose has an (energy) storage function v. cellulose has structural function / description relating to cell wall ;	2																	

Question	Answer	Marks
4(c)	<i>max 2 from mp 1 and 2 if transcription described or mix and match transcription and replication</i> <i>any five from:</i> 1 DNA (double helix / molecule) unwinds ; I unzips R DNA, strand / α helix, unwinds 2 hydrogen bonds break between, base pairs / bases / strands ; A hydrogen bonds break between nucleotides <i>only if clear that two strands are separated</i> 3 both strands used as templates ; <i>concise statement</i> 4 <u>DNA polymerase</u>, qualified ; e.g. involved in polynucleotide formation / phosphodiester bond formation / catalyses synthesis R joins phosphates 5 <i>ref. to</i> (free) activated (DNA) nucleotides / AW ; A phosphorylated nucleotides R RNA nucleotides 6 complementary (DNA) nucleotides added ; R RNA nucleotides A described in terms of complementary base pairing A A pairs with T <u>and</u> C pairs with G 7 <i>idea that</i> process, occurs / continues, along whole DNA molecule ; 8 <i>ref. to</i> Okazaki fragments / movement of polymerase in one direction / nucleotides added in one direction ; A correct <i>ref. to</i> leading and lagging strands 9 each newly formed molecule contains one original and one newly synthesised strand ; 10 AVP ; e.g. replication bubbles form / described <i>ref. to</i> repair / proofreading <i>ref. to</i> helicase (unwinding) / ligase (joining Okazaki fragments) <i>in correct context</i> R ligase joining phosphates process occurs, step-by-step / sequentially / AW <u>ref. to</u> RNA primers	5